Lab-1 CS374

Deadline: 19/08/2026

Platform: Prefer using Python or Matlab
Submission: Python - .ipynb, pdf. MATLAB - .mat/.mlx , pdf

Section A: Plotting and Function Behavior

Q1. Exploring Basic Functions
Plot the following functions on the same graph:

- (A) $y = e^x$
- (B) $y = x$
- (C) $y = \ln x$

Tasks:

- Use two ranges: $x \in [0.1, 5]$ and $x \in [-2, 2]$.
- Then plot both e^x and e^{-x} on the same graph.
- Use different scales: linear, semi-log (log-y), and log-log.

Think:

- Where do the curves intersect visually?
- How do these functions behave near $x = 0$ and as $x \to \infty$?
- What happens when you change the axes to log scale?

Q2. Gaussian and Lorentz Function Comparison
Plot the Gaussian function:

$$y = y_0 e^{-a(x-\mu)^2}$$

Fix $y_0 = 1$, and try:

- $a = 1, 2$
- $\mu = 0, 2$

Now plot the Lorentzian function:

$$y = \frac{1}{1 + a(x-\mu)^2}$$

on the same graph for the same a and μ.
Tasks:

- Compare the shape of Gaussian and Lorentz curves.
- For $x \in [0, 10]$, plot the difference between them.
- Discuss which function has a sharper peak or faster decay.

Q3. Behavior of $y = x \ln x$
Plot $y = x \ln x$ over:

- $x \in [0.01, 2]$
- $x \in [2, 20]$

Tasks:

- Plot its first derivative and interpret slope behavior.
- Examine behavior as $x \to 0^+$.
- Discuss growth rate for large x.

Optional Thought: Can this function become negative? Why or why not?

Section B: Taylor Polynomials

Q4. For the following functions:

- $y = e^x$
- $y = \ln x$ (use expansion at $a = 1$)
- $y = \sin x$
- $y = \cos x$

Tasks:

- Compute first-, second-, and third-degree Taylor polynomials.
- Plot original function and all three approximations on the same graph.
- Use expansion at $a = 0$ (except for $\ln x$, use $a = 1$).
- Also plot: $f(x) - T_n(x)$ vs x to show the error.

Q5. Construct Taylor polynomial for $\ln x$ around $a = 1$ such that:

$$|\ln(2) - T_n(2)| < 0.01$$

Tasks:

- Start with $n = 1$, and increase until the error is below 0.01.
- Report the minimum degree needed.

Q6. For $\ln x$ expanded at $a = 1$, compute Taylor polynomials from degree 1 to 10.
Tasks:

- Plot error $|\ln(x) - T_n(x)|$ vs degree n for:
  - $x = 2$, $x = 3$, $x = 4$
- Comment on how error changes with increasing n and increasing distance from a.

Section C: Floating-Point Representation and Error Propagation

Q7. Some microcomputers in the past used a binary floating-point format with 8 bits for the exponent and 1 bit for the sign. The significand contained 31 bits, with the constraint $1 \leq x < 2$, and no hiding of the leading 1. The arithmetic used rounding.

Tasks:

a) Find the machine epsilon ε.
b) Determine the number M of representable numbers between 1 and 2.
c) Give upper and lower bounds for the relative error of representing a real number using this format.

Reference: For help with this question, refer to the textbook:

- Kendall Atkinson and Weimin Han. *Elementary Numerical Analysis.* Wiley, 3rd edition, 2003.

Q8. You are given two measured quantities:

- $x = 100$ with a relative error of 1
- $y = 0.01$ with a relative error of 1

Tasks:

a) Compute the relative errors in:
   (i) $x + y$
   (ii) $x - y$
   (iii) xy
   (iv) $\frac{x}{y}$
b) Which of these expressions is more sensitive to error? Explain.
c) Consider the function:

$$f(x) = x(\sqrt{x+1} - \sqrt{x})$$

Evaluate $f(10^6)$ using both this form and:

$$f(x) = \frac{x}{\sqrt{x+1} + \sqrt{x}}$$

Compare and explain which version gives better accuracy.